Spring of Code

Artificial Intelligence

(Machine Learning & Deep Learning)

Instructor
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Duration
03 Months
(Feb – April)

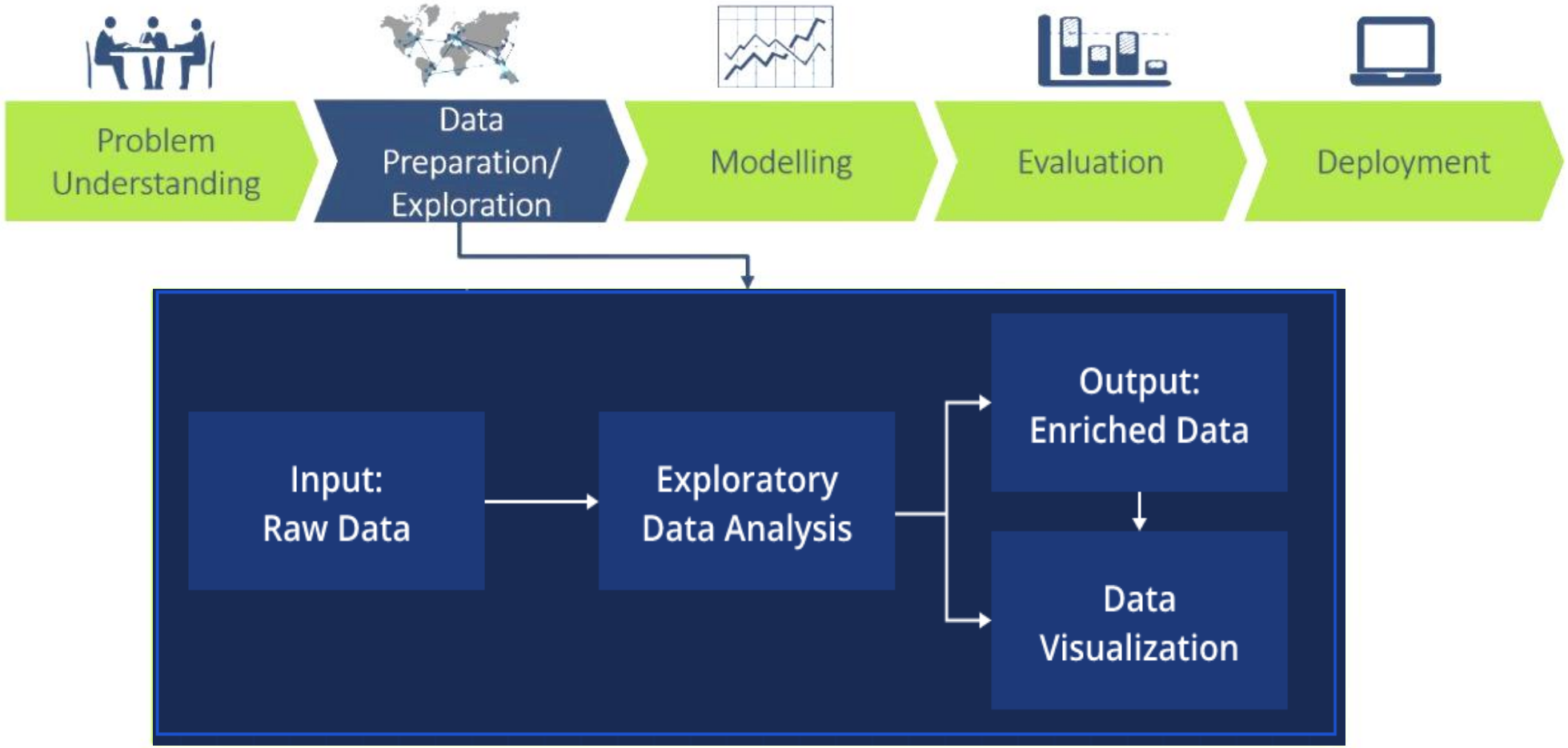
Day 02 – Exploratory Data Analysis
(Introduction to Pandas)

Objectives:

- Introduction to Pandas
- Series and DataFrame
- Operations on DataFrame
- Handling Missing Data

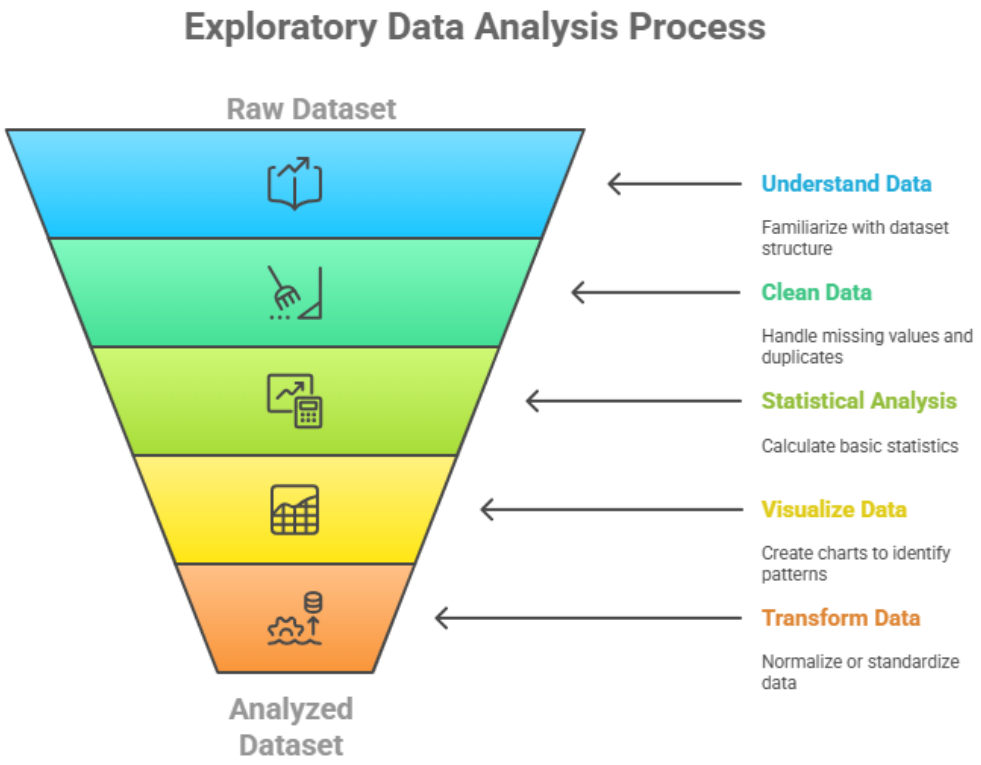
Exploratory Data Analysis

The process of examining datasets to summarize their main characteristics. It is a crucial step in the data analysis workflow to gain a deep understanding of the dataset before modeling.



Key Steps in EDA

- **Understanding the Data:** Get familiar with the dataset, check number of rows, columns, and data types.
- **Data Cleaning:** Handle missing values, duplicates, and inconsistencies.
- **Statistical Analysis:** Use basic statistics (mean, median, standard deviation) to summarize each variable.
- **Data Visualization:** Use charts to uncover patterns, trends and outliers.
- **Data Transformation** (if needed)**:** Normalize or standardize values, or convert data into a better format for further analysis or modeling.



Python Libraries for EDA

- **`NumPy:`** Essential for numerical operations in Python, it provides support for multi-dimensional arrays, along with mathematical functions on these arrays.
- **`Pandas:`** Library for data manipulation and analysis. It makes it easy to clean, transform, and aggregate data.
- **`Matplotlib:`** A versatile plotting library used to create static, interactive, and animated visualizations in Python.
- **`sklearn:`** Primarily a machine learning library but includes many tools useful for data preprocessing and feature selection, which are key parts of EDA.

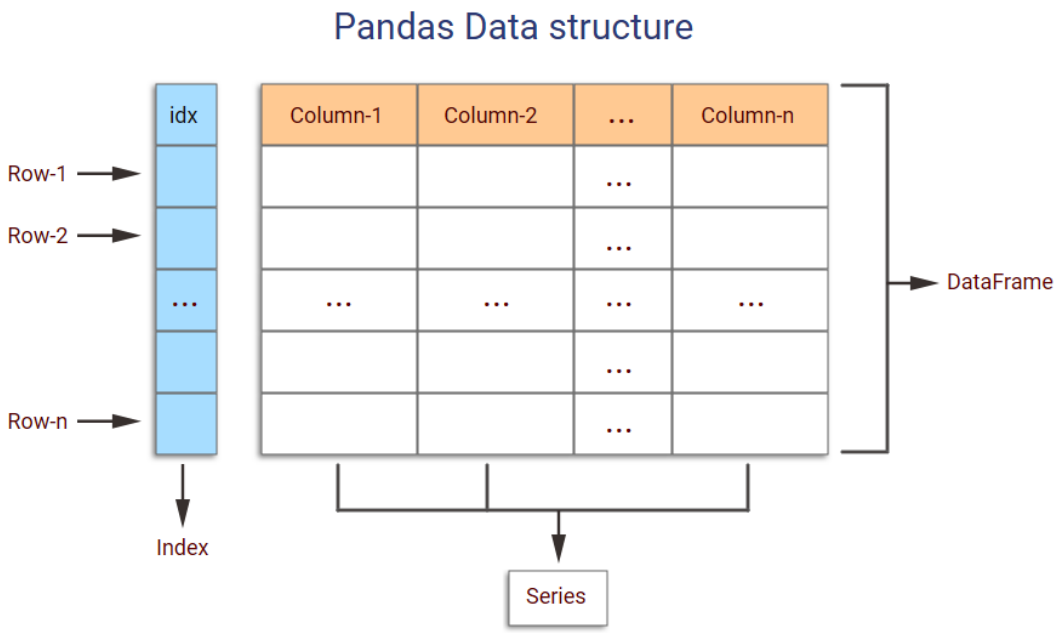
Introduction to Pandas

Pandas: The foundational library for data analysis in Python.

- **Initial release:** 2009
- Built on top of NumPy library
- **Core Data Structure**: Series and DataFrame, which are basically NumPy arrays with additional functionalities for data analysis.

Why Pandas?

- Handles structured data efficiently.
- Simplifies tasks like data cleaning, transformation, and visualization.
- Offers powerful tools for working with tabular and time-series data.




```
import pandas as pd
print(pd.__version__)
```

Pandas Data Structures

Series:

- A 1-dimensional, array-like structure with labeled indices.
- Used for storing and manipulating a single column or list of data.

DataFrame:

- A 2-dimensional tabular structure with rows and columns.
- Can be created from dictionaries, lists, or NumPy arrays.

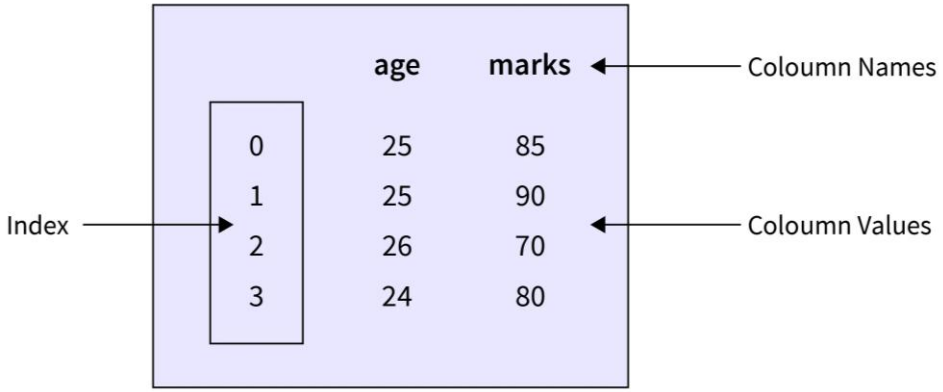
Key Features:

- Supports heterogeneous data types.
- Easy data manipulation and aggregation.
- Offers methods to filter, group, and transform data efficiently.

Series

Index	Data
0	Mark
1	Justin
2	John
3	Vicky

DataFrame



Pandas Series

Variable name

To which we store the Series

Name

The name of the Series (optional). None by default

s

	C1
0	A
1	B
2	C

Index

The default is for these to be integers 0,1,2,… However, you can set them manually using the "index" keyword

Data

The data of the Series. Can be of almost any type you need to represent your data including strings, integers, floats, dates, Booleans, and more.

```
s = pd.Series(data=["A","B","C"],
              name="C1")
```

Pandas DataFrame

SERIES DATA INDICES LABELS AXIS

	0	1	2	3	4	5
	homeTeamName	awayTeamName	startTime	duration_minutes	attendance	dayNight
0	Cubs	Reds	2016-07-05 18:20:00 UTC	188	41310	D
1	Indians	Astros	2016-09-08 16:10:00 UTC	194	15275	D
2	Padres	Giants	2016-09-25 20:40:00 UTC	185	28456	D
3	Diamondbacks	Brewers	2016-08-07 20:10:00 UTC	211	24021	D
4	Giants	Dodgers	2016-04-09 20:05:00 UTC	202	41224	D
5	Blue Jays	Indians	2016-07-02 17:07:00 UTC	199	46197	D
6	Reds	Pirates	2016-04-09 17:10:00 UTC	180	22799	D
7	Cubs	Mariners	2016-07-30 18:20:00 UTC	157	41401	D
8	Rockies	Phillies	2016-07-10 20:10:00 UTC	191	32113	D
9	Yankees	Blue Jays	2016-05-26 20:05:00 UTC	154	38391	D

INDICES

AXIS

Indexing and Selection

.iloc selections - position based selection

```
data.iloc[<row selection], <column selection>]
```

Integer list of rows: [0,1,2]
Slice of rows: [4:7]
Single values: 1

Integer list of columns: [0,1,2]
Slice of columns: [4:7]
Single column selections: 1

---

loc selections - position based selection

```
data.loc[<row selection], <column selection>]
```

Index/Label value: 'john'
List of labels: ['john', 'sarah']
Logical/Boolean index: data['age'] == 10

Named column: 'first_name'
List of column names: ['first_name', 'age']
Slice of columns: 'first_name':'address'

Pandas Basic Operations

Pandas simplifies data handling by enabling efficient preprocessing, cleaning, transformation, and visualization.

01 DATA PREPROCESSING

Handle missing values

02 DATA CLEANING

Remove duplicates and fix inconsistencies

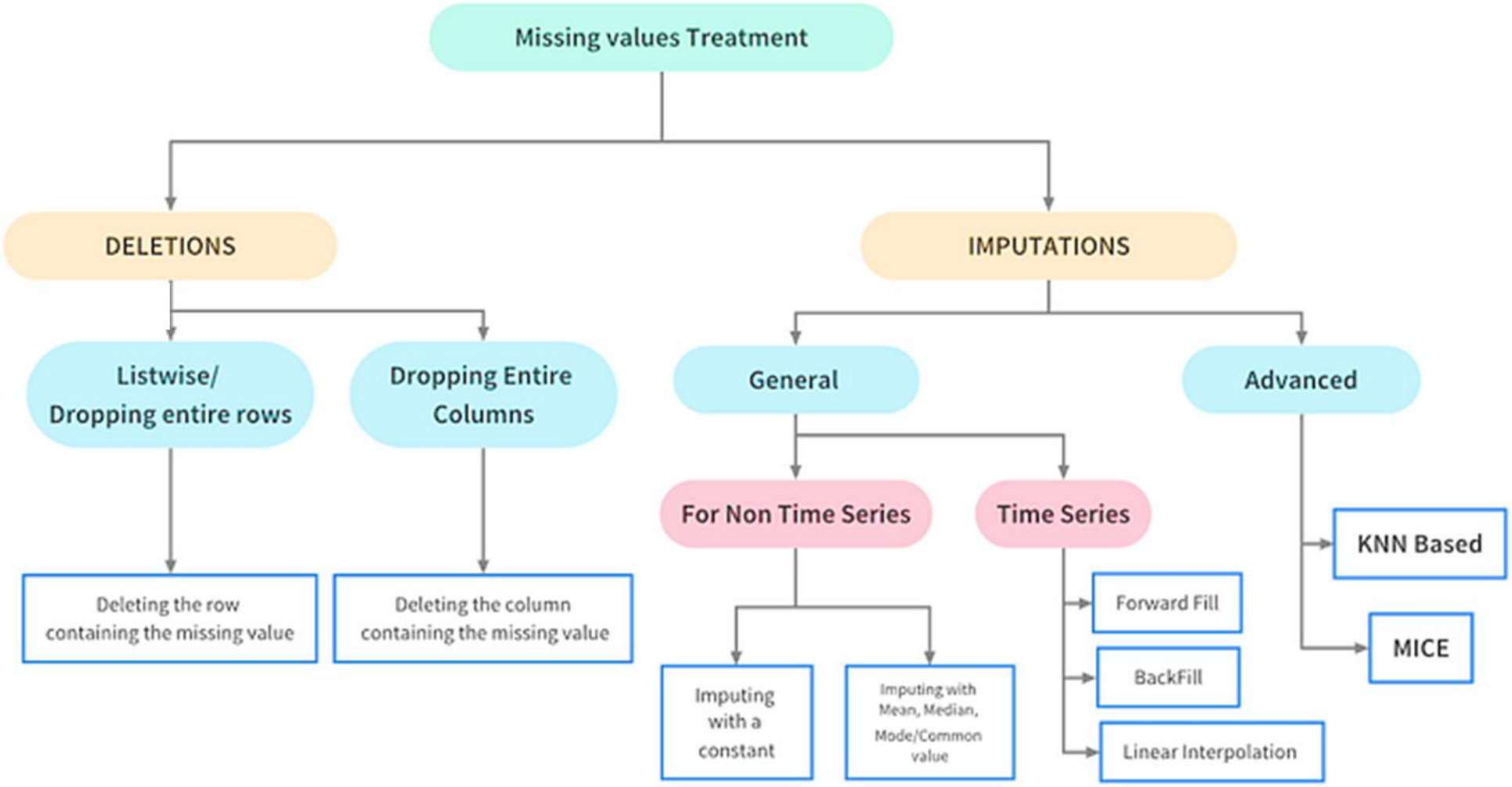
03 DATA TRANSFORMATION

Filter, sort & modify data

04 DATA VISUALIZATION

Represent data graphically

Handling Missing Values



Thank You